

UBUNTU AWS SERVER CONFIGURATION MANUAL

Update/Upgrade package lists

```
sudo apt update
sudo apt upgrade -y
sudo reboot
```

Set Timezone

```
timedatectl
sudo timedatectl set-timezone Africa/Nairobi
[verify]-( Time zone: Africa/Nairobi (EAT))
```

Set Hostname

```
sudo hostnamectl set-hostname inventory-server
[verify]- hostnamectl
```

Install Essential Server Tools

```
sudo apt update
sudo apt install -y curl wget git nano vim zip unzip tree htop btop net-tools software-properties-
common ca-certificates gnupg lsb-release apt-transport-https cron ufw fail2ban openssl build-essential
rsync bash-completion dnsutils acl jq
[Verify the installation] - git --version, curl --version, htop --version, btop --version, ufw --version,
fail2ban-client --version
```

Configure UFW Firewall

```
[Allow SSH] - sudo ufw allow OpenSSH
[Allow HTTP] - sudo ufw allow 80/tcp
[Allow HTTPS] - sudo ufw allow 443/tcp
[Enable firewall] - sudo ufw enable
[Check status] - sudo ufw status
```

Configure Lightsail Firewall

Networking > Allow > Ports 22, 80, 443

Create Static IP

Lightsail > Networking > Static IP > Create Static IP > Attach to the instance

Enable Automatic Security Updates

```
[Install] - sudo apt install unattended-upgrades -y
[Enable] - sudo dpkg-reconfigure unattended-upgrades > Yes
```

Verify Server Health

```
free -h
df -h
lscpu
ip addr
```

Update/Upgrade package lists

```
sudo apt update
```

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sudo apt upgrade -y

Install Apache

sudo apt install apache2 -y

[Enable Apache] - sudo systemctl enable apache2

[Start Apache] - sudo systemctl start apache2

[Check status] - sudo systemctl status apache2 (Active: active (running))

[Verify Apache] - http://YOUR_STATIC_IP

Check the PHP Version Available

apt policy php

Install PHP and Extensions

sudo apt install -y php php-cli php-common php-fpm php-mysql php-curl php-gd php-mbstring php-xml php-zip php-bcmath php-intl php-soap php-imap php-readline php-opcache php-redis php-imagick libapache2-mod-php

[Verify PHP] - php -v

[Loaded Modules] - php -m

Enable PHP Production Settings

sudo nano /etc/php/8.3/apache2/php.ini

[Search **Ctrl + W**)

memory_limit = 512M

upload_max_filesize = 100M

post_max_size = 100M

max_execution_time = 300

max_input_time = 300

date.timezone = Africa/Nairobi

display_errors = Off

log_errors = On

max_input_vars = 5000

session.cookie_httponly = On

session.use_strict_mode = On

expose_php = Off

Install MariaDB

sudo apt install mariadb-server mariadb-client -y

[Enable] - sudo systemctl enable mariadb

[Start] - sudo systemctl start mariadb

[Check status] - sudo systemctl status mariadb (Active: active (running))

Secure MariaDB

sudo mysql_secure_installation

Enter current password for root: Press Enter

Switch to unix_socket authentication? Y

Change the root password? Y

Remove anonymous users? Y

Disallow root login remotely? Y

Remove test database? Y

Reload privilege tables? Y

Login: sudo mysql

[Check Version] - SELECT VERSION();

EXIT;

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Enable Apache Modules

sudo a2enmod rewrite headers expires ssl

[Restart] - sudo systemctl restart apache2

Test PHP

echo "<?php phpinfo(); ?>" | sudo tee /var/www/html/info.php

[Test] - http://YOUR_SERVER_IP/info.php

{Delete file after testing} – sudo rm /var/www/html/info.php

Create a Database

sudo mysql

CREATE DATABASE inventory_db CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;

[Create User]

CREATE USER 'inventory_user'@'localhost' IDENTIFIED BY 'StrongPassword123!';

[Grant Database Permissions]

GRANT ALL PRIVILEGES ON inventory_db.* TO 'inventory_user'@'localhost';

FLUSH PRIVILEGES;

[Cofirm] - SHOW DATABASES;

EXIT;

Go to web root

ls -la

[Remove apache default page] - sudo rm -f /var/www/html/index.html

[Clone github Repo] - sudo git clone https://github.com/YOUR_USERNAME/inventory_system.git

[Verify]

Change Ownership

sudo chown -R www-data:www-data /var/www/html/inventory_system

Set Permissions

[Directories] - sudo find /var/www/html/inventory_system -type d -exec chmod 755 {} \;

[Files] - sudo find /var/www/html/inventory_system -type f -exec chmod 644 {} \;

Give Write Permission to Required Folders

sudo chmod -R 775 /var/www/html/inventory_system/uploads

sudo chmod -R 775 /var/www/html/inventory_system/logs

Import Database

Go to database file Location

Sudo mysql -u inventory_user -p inventory_db < inventory_db.sql

[Verify] - sudo mysql

Create the Apache Virtual Host

sudo nano /etc/apache2/sites-available/inventory.mombasacomputers.com.conf

[Content]

<VirtualHost *:80>

ServerAdmin admin@mombasacomputers.com

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ServerName inventory.mombasacomputers.com

DocumentRoot /var/www/html/inventory_system

<Directory /var/www/html/inventory_system>

AllowOverride All

Require all granted

</Directory>

ErrorLog \${APACHE_LOG_DIR}/inventory_error.log

CustomLog \${APACHE_LOG_DIR}/inventory_access.log combined

</VirtualHost>

[Enable site]

sudo a2ensite inventory.mombasacomputers.com.conf

[Disable default site] - sudo a2dissite 000-default.conf

[Test configuration]

sudo apache2ctl configtest (Syntax OK)

[Restart] sudo systemctl restart apache2

Create the DNS Record

[Verify] - nslookup inventory.mombasacomputers.com/ dig inventory.mombasacomputers.com

[Test] - curl -I <http://domain> (HTTP/1.1 200 OK)

Install Certbot

sudo apt update

sudo apt install -y certbot python3-certbot-apache

[Verify] - certbot --version

Check apache Virtual

sudo apache2ctl -S

Enable SSL and Rewrite modules

sudo a2enmod ssl

sudo a2enmod rewrite

sudo a2enmod headers

[Restart] - sudo systemctl restart apache2

[Check] - sudo apache2ctl configtest (Syntax OK)

Obtain the SSL certificate

sudo certbot --apache -d inventory.mombasacomputers.com

[Enter Email address]

[Redirect HTTP Traffic to HTTPS]

[Check apache configuration] - sudo apache2ctl -S

(*:80, *:443)

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Look at the HTTPS virtual host

```
sudo cat /etc/apache2/sites-available/inventory.mombasacomputers.com.conf
```

```
<IfModule mod_ssl.c>
```

```
<VirtualHost *:443>
```

```
    ServerAdmin admin@mombasacomputers.com
```

```
    ServerName inventory.mombasacomputers.com
```

```
    DocumentRoot /var/www/html/inventory_system
```

```
    <Directory /var/www/html/inventory_system>
```

```
        AllowOverride All
```

```
        Require all granted
```

```
    </Directory>
```

```
    ErrorLog ${APACHE_LOG_DIR}/inventory_error.log
```

```
    CustomLog ${APACHE_LOG_DIR}/inventory_access.log combined
```

```
    SSLEngine on
```

```
    SSLCertificateFile /etc/letsencrypt/live/inventory.mombasacomputers.com/fullchain.pem
```

```
    SSLCertificateKeyFile /etc/letsencrypt/live/inventory.mombasacomputers.com/privkey.pem
```

```
</VirtualHost>
```

```
</IfModule>
```

Test apache

```
sudo apache2ctl configtest – (Syntax OK)
```

```
[Reload] - sudo systemctl reload apache2
```

Test HTTPS

```
curl -I https://inventory.mombasacomputers.com - (Http Response)
```

```
curl -I http://inventory.mombasacomputers.com – (HTTP/1.1 301 Moved Permanently
```

```
Location: https://inventory.mombasacomputers.com/)
```

Check Certificate

```
sudo certbot certificates
```

Test automatic certificate renewal

```
sudo certbot renew --dry-run
```

```
[Check the timer] - systemctl list-timers | grep certbot
```

Verify Apache is listening on HTTPS

```
sudo ss -tulpn | grep ':443'
```

```
sudo ss -tulpn | grep ':80'
```

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Install phpMyAdmin

sudo apt update

sudo apt install -y phpMyAdmin

Configure database for phpmyadmin with dbconfig-common? – YES

[Verify Installation] - ls -la /usr/share/phpMyAdmin (index.php, libraries, templates, config.inc.php)

[Check apache config] - sudo apache2ctl configtest (Syntax OK)

[Restart]

[Check apache loaded phpMyAdmin] - ls -la /etc/apache2/conf-enabled/ | grep phpMyAdmin
(phpmyadmin.conf)

[Enable] - sudo a2enconf phpMyAdmin

[Reload]

[Apache auth utilities] - sudo apt install -y apache2-utils

[Dedicated phpMyAdmin admin] - sudo htpasswd -c /etc/apache2/.phpmyadmin_htpasswd dbadmin

Protect phpMyAdmin with Apache

[Backup] - sudo cp /etc/apache2/conf-available/phpmyadmin.conf /etc/apache2/conf-available/phpmyadmin.conf.backup

[Config] - sudo nano /etc/apache2/conf-available/phpmyadmin.conf

/usr/share/phpMyAdmin ->:

<Directory /usr/share/phpmyadmin>

Options SymLinksIfOwnerMatch

DirectoryIndex index.php

AllowOverride All

AuthType Basic

AuthName "Restricted Database Administration"

AuthUserFile /etc/apache2/.phpmyadmin_htpasswd

Require valid-user

</Directory>

Confirm apache supports auth

sudo a2enmod auth_basic

sudo apache2ctl configtest

(Syntax OK)

[Reload]

Verify db is not publicly accessible

sudo ss -lntp | grep 3306

sudo ufw status numbered (sudo ufw delete allow 3306/tcp)